

# KCC Platform & PlenorHub — Server Administration & Operations Guide

This document is the operational manual for managing, updating, and administering the KCC Platform and PlenorHub server infrastructure.

## 1. Server Architecture & Directory Layout

### Server Details

- **Host:** Tencent Cloud Lighthouse (CentOS Stream 8)
- **Public IP:** 43.156.19.185
- **Control Panel:** aaPanel (runs alongside system services)

### Live Application Matrix

| Service                   | Domain                         | Internal Path                   | Tech Stack           |
|---------------------------|--------------------------------|---------------------------------|----------------------|
| <b>KCC ID Auth API</b>    | https://auth.bigkpay.com       | /var/www/kccid-api              | Laravel 11 / PHP 8.2 |
| <b>BigK Admin API</b>     | https://api.bigkpay.com        | /var/www/bigk-api/backend       | Laravel 11 / PHP 8.2 |
| <b>PlenorHub API</b>      | https://api.plenorhub.com      | /var/www/plenorhub-api/backend  | Laravel 11 / PHP 8.2 |
| <b>BigK Admin Panel</b>   | https://admin.bigkpay.com      | /var/www/bigk-admin-web         | Flutter Web static   |
| <b>BigK Wallet</b>        | https://app.bigkpay.com        | /var/www/bigk-wallet-web        | Flutter Web static   |
| <b>KCC ID Dashboard</b>   | https://id.kccdigital.com      | /var/www/kccid-dashboard        | Flutter Web static   |
| <b>PlenorHub Merchant</b> | https://merchant.plenorhub.com | /var/www/plenorhub-merchant-web | Flutter Web static   |
| <b>PlenorHub Admin</b>    | https://admin.plenorhub.com    | /var/www/plenorhub-admin-web    | Flutter Web static   |

### Key Server Infrastructure Components

#### 1. PHP 8.2 FPM

- **Binary Path:** /www/server/php/82/bin/php
- **Composer Path:** /usr/local/bin/composer
- **FPM Socket:** /tmp/php-cgi-82.sock
- **Service Control:** /etc/init.d/php-fpm-82 restart

## 2. PostgreSQL Database Engine

- **Host:** 127.0.0.1 (Port 5432)
- **Databases:**
  - kccid (User: kccid\_user)
  - bigk\_admin (User: bigk\_user)
  - plenorhub (User: plenorhub\_user)

## 3. Supervisor (Background Queue Workers)

- **Config Directory:** /etc/supervisord.d/
- **Active Workers:**
  - kccid-worker (/var/log/kccid-worker.log)
  - bigk-worker (/var/log/bigk-worker.log)
  - plenorhub-worker (/var/log/plenorhub-worker.log)
- **Control Command:** supervisorctl -c /etc/supervisord.conf status

## 4. Nginx Web Server

- **Vhost Directory:** /www/server/panel/vhost/nginx/
  - **Binary Path:** /www/server/nginx/sbin/nginx
  - **Reload Command:** systemctl reload nginx
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## 2. GitHub Integration & Code Update Workflows

### GitHub Account & Organization Setup (kccdigital)

All production repositories belong to the kccdigital GitHub organization: -  
github.com/kccdigital/kccid - github.com/kccdigital/bigk-admin -  
github.com/kccdigital/bigk-wallet - github.com/kccdigital/plenorhub

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### Generating a GitHub Personal Access Token (PAT)

To allow the Tencent server to pull private repositories from kccdigital, generate a Personal Access Token:

1. Log into your **GitHub account** under kccdigital.
2. Go to **Settings** → **Developer Settings** → **Personal Access Tokens** → **Tokens (classic)**.
3. Click **Generate new token (classic)**.
4. Set Note to Tencent Production Server and select scopes:

- repo (Full control of private repositories)
5. Click **Generate token** and copy the token string (starts with ghp\_ or github\_pat\_).
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## Connecting Server Repositories to kccdigital

Run these commands on the server terminal to set or update the remote URL for each application:

*# 1. KCC ID Auth Backend*

```
cd /var/www/kccid-api
git remote set-url origin
https://<YOUR_GITHUB_PAT>@github.com/kccdigital/kccid.git
```

*# 2. BigK Admin Backend*

```
cd /var/www/bigk-api
git remote set-url origin
https://<YOUR_GITHUB_PAT>@github.com/kccdigital/bigk-admin.git
```

*# 3. PlenorHub Backend*

```
cd /var/www/plenorhub-api
git remote set-url origin
https://<YOUR_GITHUB_PAT>@github.com/kccdigital/plenorhub.git
```

Verify that remotes are configured correctly:

```
cd /var/www/kccid-api && git remote -v
cd /var/www/bigk-api && git remote -v
cd /var/www/plenorhub-api && git remote -v
```

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## Step-by-Step: Updating Backend Code

When code changes or database migrations are pushed to GitHub, run these commands on the server:

*KCC ID Auth Backend (/var/www/kccid-api)*

```
cd /var/www/kccid-api
git stash 2>/dev/null
git pull origin main
/www/server/php/82/bin/php /usr/local/bin/composer install --no-dev
--optimize-autoloader
/www/server/php/82/bin/php artisan migrate --force
/www/server/php/82/bin/php artisan config:clear
/www/server/php/82/bin/php artisan cache:clear
/www/server/php/82/bin/php artisan route:clear
chown -R www:www storage bootstrap/cache
systemctl restart php-fpm-82
supervisorctl -c /etc/supervisord.conf restart kccid-worker
```

#### *BigK Admin Backend (/var/www/bigk-api)*

```
cd /var/www/bigk-api
git stash 2>/dev/null
git pull origin main
cd backend
/www/server/php/82/bin/php /usr/local/bin/composer install --no-dev
--optimize-autoloader
/www/server/php/82/bin/php artisan migrate --force
/www/server/php/82/bin/php artisan config:clear
/www/server/php/82/bin/php artisan cache:clear
/www/server/php/82/bin/php artisan route:clear
chown -R www:www storage bootstrap/cache
systemctl restart php-fpm-82
supervisorctl -c /etc/supervisord.conf restart bigk-worker
```

#### *PlenorHub Backend (/var/www/plenorhub-api)*

```
cd /var/www/plenorhub-api
git stash 2>/dev/null
git pull origin main
cd backend
/www/server/php/82/bin/php /usr/local/bin/composer install --no-dev
--optimize-autoloader
/www/server/php/82/bin/php artisan migrate --force
/www/server/php/82/bin/php artisan config:clear
/www/server/php/82/bin/php artisan cache:clear
/www/server/php/82/bin/php artisan route:clear
chown -R www:www storage bootstrap/cache
systemctl restart php-fpm-82
supervisorctl -c /etc/supervisord.conf restart plenorhub-worker
```

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### **Step-by-Step: Deploying Web Frontends**

Frontend web apps are compiled locally using Flutter, pushed to GitHub, and copied into public web roots on the server.

#### *1. Local Machine Build & Push*

##### *# BigK Admin Panel*

```
cd ~/Desktop/repos/bigk-admin/frontend
flutter build web --release
git add -f build/web
git commit -m "deploy: updated admin frontend build"
git push origin main
```

##### *# BigK Wallet App*

```
cd ~/Desktop/repos/bigk-wallet
dart run build_runner build --delete-conflicting-outputs
flutter build web --release
git add -f build/web
```

```
git commit -m "deploy: updated wallet build"
git push origin main
```

#### *# KCC ID Dashboard*

```
cd ~/Desktop/repos/kccid/dashboard
flutter build web --release
--dart-define=API_BASE_URL=https://auth.bigkpay.com
cd ..
git add -f dashboard/build/web
git commit -m "deploy: updated dashboard build"
git push origin main
```

#### *# PlenorHub Merchant & Admin Panels*

```
cd ~/Desktop/repos/plenorhub/frontend
flutter build web --release --dart-define=APP_MODE=merchant
cp -r build/web build/web-merchant
flutter build web --release --dart-define=APP_MODE=admin
cp -r build/web build/web-admin
cd ..
git add -f frontend/build/web-merchant frontend/build/web-admin
git commit -m "deploy: updated merchant and admin builds"
git push origin main
```

## *2. Server Deployment Commands*

Run these on the server terminal to deploy built assets:

#### *# Deploy BigK Admin Web*

```
cd /var/www/bigk-api && git pull origin main
\cp -rf frontend/build/web/* /var/www/bigk-admin-web/
```

#### *# Deploy BigK Wallet Web*

```
rm -rf /tmp/bigk-wallet-tmp
git clone https://<YOUR_GITHUB_PAT>@github.com/kccdigital/bigk-wallet.git
/tmp/bigk-wallet-tmp
\cp -rf /tmp/bigk-wallet-tmp/build/web/* /var/www/bigk-wallet-web/
```

#### *# Deploy KCC ID Dashboard*

```
cd /var/www/kccid-api && git pull origin main
\cp -rf dashboard/build/web/* /var/www/kccid-dashboard/
```

#### *# Deploy PlenorHub Webfronts & Landing*

```
cd /var/www/plenorhub-api && git pull origin main
\cp -rf frontend/build/web-merchant/* /var/www/plenorhub-merchant-web/
\cp -rf frontend/build/web-admin/* /var/www/plenorhub-admin-web/
\cp -rf landing/* /var/www/plenorhub-landing/
```

#### *# Set Web Permissions*

```
chown -R www:www /var/www/bigk-admin-web /var/www/bigk-wallet-web
```

```
/var/www/kccid-dashboard /var/www/plenorhub-merchant-web  
/var/www/plenorhub-admin-web /var/www/plenorhub-landing
```

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### 3. PlenorHub Channel Management & API Keys

PlenorHub serves products to multiple apps (bigk, kincore, shuyi, numforlife). Catalog distribution is controlled by **Channel API Keys**.

#### Minting a New Channel API Key

To add a new ecosystem app or issue a new key:

```
cd /var/www/plenorhub-api/backend  
PHP=/www/server/php/82/bin/php  
  
# Mint key for existing or new channel key (e.g. kincore)  
$PHP artisan channel:create-credential kincore --name="Kincore Production  
Key"
```

The terminal will print the plaintext api\_key (starts with ph\_ch...). Store this key securely; it is hashed in the database and cannot be retrieved again.

#### Viewing Active Channel Credentials

```
cd /var/www/plenorhub-api/backend  
/www/server/php/82/bin/php artisan tinker --execute="  
\App\Models\Mall\ChannelCredential::with('channel')->get(['id', 'channel_id',  
'name', 'status', 'created_at'])->each(function(\$c) {  
    echo \"ID: {\$c->id} | Channel: {\$c->channel->channel_key} | Name:  
{\$c->name} | Status: {\$c->status}\" . PHP_EOL;  
});  
"
```

#### Rotating or Revoking a Channel Key

If a key is compromised or needs rotation:

1. Mint a new key using channel:create-credential.
2. Update the calling application with the new key.
3. Revoke the old credential ID:

```
cd /var/www/plenorhub-api/backend  
/www/server/php/82/bin/php artisan tinker --execute="  
\App\Models\Mall\ChannelCredential::where('id',  
<CREDENTIAL_ID>)->update(['status' => 'revoked']);  
echo 'Credential revoked.' . PHP_EOL;  
"
```

#### Enforcing Channel Authentication Policy

To enforce channel API key checks on /integration/\* endpoints:

1. Open `/var/www/plenorhub-api/backend/.env`
2. Set `INTEGRATION_REQUIRE_CHANNEL_AUTH=true`
3. Clear config cache:

```
cd /var/www/plenorhub-api/backend
/www/server/php/82/bin/php artisan config:clear
```

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## 4. Credential Management & Swapping Guide

All environment configurations are stored in `.env` files directly on the server.

### How to Edit `.env` Files and Apply Changes

To modify environment variables:

```
# Open editor for specific backend
nano /var/www/kccid-api/.env           # KCC ID
nano /var/www/bigk-api/backend/.env    # BigK Admin
nano /var/www/plenorhub-api/backend/.env # PlenorHub
```

After saving edits (Ctrl+O, Enter, Ctrl+X), clear the config cache and restart PHP-FPM:

```
cd /var/www/<project-path>
/www/server/php/82/bin/php artisan config:clear
/www/server/php/82/bin/php artisan cache:clear
systemctl restart php-fpm-82
```

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### Admin account password change through the server CLI Reset via KCC ID

Run this command on the server terminal inside the `/var/www/kccid-api` directory:

```
cd /var/www/kccid-api
/www/server/php/82/bin/php artisan tinker --execute="
\$user = \App\Models\KccId\KccUser::where('email',
'admin@example.com')->first();
if (\$user) {

app(\App\Services\KccId\IdentityService::class)->updatePasswordAndInvalidatesS
essions(\$user->kcc_user_id, 'NewSecurePassword123!');
    echo 'KCC ID Admin password updated successfully. Existing sessions
signed out.' . PHP_EOL;
} else {
    echo 'Admin user not found in KCC ID.' . PHP_EOL;
```

```
}  
"
```

Replace 'admin@example.com' with the target admin's email and 'NewSecurePassword123!' with the new password.

## 5. Server Health Checks & Troubleshooting Cheat Sheet

### Server Health Command Checklist

Run these commands to verify operational status across all components:

```
# 1. Check HTTP status of all live domains  
for url in https://auth.bigkpay.com https://api.bigkpay.com  
https://api.plenorhub.com https://admin.bigkpay.com https://app.bigkpay.com  
https://id.kccdigital.com https://plenorhub.com  
https://merchant.plenorhub.com https://admin.plenorhub.com; do  
    echo "$url -> $(curl -s -o /dev/null -w '%{http_code}' $url)"  
done
```

```
# 2. Check Queue Worker Status  
supervisorctl -c /etc/supervisord.conf status
```

```
# 3. Check System Memory and Disk Usage  
free -h  
df -h /
```

---

### Log Inspection Commands

When investigating issues or errors:

```
# Laravel Application Logs  
tail -n 100 /var/www/kccid-api/storage/logs/laravel.log  
tail -n 100 /var/www/bigk-api/backend/storage/logs/laravel.log  
tail -n 100 /var/www/plenorhub-api/backend/storage/logs/laravel.log
```

```
# Nginx Error Log  
tail -n 100 /www/wwwlogs/nginx_error.log
```

```
# PHP-FPM Log  
tail -n 100 /www/server/php/82/var/log/php-fpm.log
```

```
# Worker Logs  
tail -n 100 /var/log/kccid-worker.log  
tail -n 100 /var/log/bigk-worker.log  
tail -n 100 /var/log/plenorhub-worker.log
```

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## Troubleshooting Common Issues

### *Issue 1: "500 Internal Server Error" after code update*

**Cause:** Folder permissions on storage or bootstrap/cache.

**Fix:**

```
cd /var/www/<project-path>
chown -R www:www storage bootstrap/cache
chmod -R 775 storage bootstrap/cache
/www/server/php/82/bin/php artisan config:clear
/www/server/php/82/bin/php artisan cache:clear
/etc/init.d/php-fpm-82 restart
```

### *Issue 2: Web app displays old version after deployment*

**Cause:** Browser ServiceWorker and HTTP caching.

**Fix:** perform a hard refresh (Ctrl+Shift+R or Cmd+Shift+R).

### *Issue 3: SSL Certificate expiration or domain mismatch*

**Fix:** Re-run certbot for the domain group:

```
certbot --nginx \
  --nginx-ctl /www/server/nginx/sbin/nginx \
  --nginx-server-root /www/server/nginx/conf \
  -d plenorhub.com -d merchant.plenorhub.com -d admin.plenorhub.com \
  --non-interactive --agree-tos --email munene010@gmail.com
```

```
systemctl reload nginx
```

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## Emergency Service Restart Sequence

If multiple services become unresponsive, execute this complete restart sequence:

```
/etc/init.d/php-fpm-82 restart
systemctl reload nginx
supervisorctl -c /etc/supervisord.conf restart all
```